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High-level approach for generating household data, dialogs, and extraction metrics

This page summarizes the end-to-end pipeline covering data generation, dialog generation, extraction scoring, Finetuned-model fine-tuning, and evaluation.

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Synthetic households	Generated dialogs	Extract and score	Finetuned-model eval
5,000	316	99	98.74%
Generated with seed 42	Train and dev splits combined	Test split only	Held-out test split

Overview · Data generation · Dialog generation · Metrics · Train and holdout test · Results · Quality control · Next steps · Artifacts

Overview

The benchmark is built and controlled synthetic-data pipeline

1. **Generate households** — Seed-reproducible households from explicit scenario distributions, with optional corruption.
2. **Generate dialogs** — Llama 3.3 70B via vLLM, OpenAI-compatible provider, or a rule-based fallback, with validator-gated retry for field-level hallucinations.
3. **Extract and score** — Run structured extraction from transcripts and score against ground truth. Scores overall, per-scenario, and per-field accuracy.
4. **Validate holdout test** — Confirm a held-out test set that was not used while tuning the extractor or evaluator.

Coverage requirement	Affirmative confirmation	Validation
11 scenarios	83% dialogs	0 rule violations
Every household must have at one named coverage scenario.	Assistant must explicitly confirm the household's profile, with relevant fields reflected.	Rule-based validator checks transcripts against ground truth. Failures trigger a bounded retry.

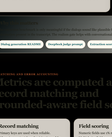
Data generation

Household data is scenario-driven and distributionally checked

The generator creates sufficient US address-like households rather than a geographically realistic address dataset. Distributions are explicit and validated for coverage. Public-data anchors and external geocoding remain future work. Synthetic household structure, not realism, is the primary goal.

Scenario coverage in the 5,000-household synthetic population

Scenario	Count	Share
family_with_partner_and_children	898	17.96%
different_single_g_female_gen_neutral	896	17.92%
pro_retirement_security	535	10.70%
one_high_earner_one_low_earner	512	10.24%
young_dual_income_no_children	418	8.36%
self_employed_business_owner	408	8.12%
financially_stressed_with_debt	401	8.02%
retiree_needs_high_guaranteed_income	401	8.02%
secondary_wedded_pairing_of_elderly	257	5.14%
divorced	215	4.30%
widowed	200	3.99%



Scenario coverage confirms that all 11 target scenarios are represented.

Household annual gross income shows a broad distribution, with most households in the low-to-mid income range and a long tail up to high-income households.

Investable assets distribution, generated as a noisy income-linked wealth model.

Income vs investable assets shows the noisy link between income and investable assets.

Risk tolerance distribution is centered on a categorical prior rather than a single mean.

Conditional mortgage tendency by scenario checks dependence structure.

Dialog generation

Dialogs are generated in stages and checked for grounding and realism

Dialog generation does not ask a model to produce the entire transcript in one shot. It builds a persona, an outline, plans local dialog turns, and then renders the transcript. Validators and retries exist where the model has a measurable failure mode.

Grounded transcript construction

The dialog generator receives the household profile and a structured outline, then renders a transcript under validation checks. Failed outputs are rejected and retried with validator-aware feedback, and plain-text transcript files.

Evidence anchors keep dialog details tied to structured facts. These evidence fields later become dialog-grounded GT for extraction scoring.

Deep/Rock realism gate

A rule-based gate classifies each transcript as a 'rock' or 'deep' conversation. Rock transcripts are kept as realistic but simple cases; deep transcripts are the challenging cases.

In the Train set, gate drop rates are reported per scenario. For the holdout test, the reported evaluation uses an equal-generated dialog sub-sample to keep authoring balanced.

Dialog generation 83% affirm | Deep/Rock judge prompt | Extraction scoring details

Scoring and metrics

Metrics are computed after record matching and grounded-aware field scoring

Extraction is evaluated against ground truth. Record matching aligns predicted records to ground truth, then field-level accuracy, coverage, and error counts are computed. Dialog-grounded GT is kept separate from strict schema GT so grounded fields can be scored without forcing unsupported extractions into the strict target.

Record matching

Primary keys are used when available. Otherwise, content-based pairing is used for people, income, debts, assets, liabilities, and pension policies.

Field scoring

Numeric fields use a tolerance. Enums, booleans, and date fields are compared as-is. Missing fields are counted separately from incorrect extractions.

Error types

Common error types include missing extracted values, invented extra values, value mismatches, and record pairing failures.

Accuracy	Implementation
Dialog-grounded denominator	When `ground_truth_is_grounded_only` only fields present in grounded GT enter the denominator.
Enum normalization	Primary keys and `_id` fields are excluded by default.
Optional ratio normalization	Fields ending in `_ratio` are excluded.
Pension-specific normalization	Configured in `ALL_PENSION_NORMALIZATIONS` and `PENSION_PRIMARY_KEYS` in `scoring/normalizers.py`.

Why it matters
Extractor is penalized for facts never stated in the dialog.
We evaluate semantic extraction, not synthetic key guessing.
Pension fields are derived and would otherwise double-count numeric facts.
Reduces noise from fields that are not central to financial extraction quality.

Train and holdout

Train was used for pipeline development; holdout test is a held-out evaluation

Split summary

Split	Households	Generated GT points	Median income	P10/P90 income	Median assets	P10/P90 assets	Ratio
Train	316	1,138	$55,125	$36,386 / $126,454	$55,000	$300 / $1,000,000	Pipeline debugging and tuning
Holdout Test	99	355	$121,988	$71,450 / $271,454	$63,750	$0 / $1,000,000	Held-out evaluation

Results

Train and holdout test performance are closely aligned

Overall metrics

Metric	Train (316)	Holdout test (99)
Record-level F1	98.78%	98.74%
Exact record rate	96.73%	96.73%
Dialogs with 100% correct fields	297 / 316 = 93.9%	92 / 99 = 92.9%
Dialogs with at least 90% correct fields	298 / 316 = 93.7%	93 / 99 = 93.9%
Dialogs with at least 80% correct fields	302 / 316 = 95.6%	97 / 99 = 98.0%
Missing extracted rows	70	30
Invented extra rows	0	0
Value mismatch rows	164	58

Train per-entity metrics

Entity	Missed	Errors	Invented	GT
people	0.3%	0.3%	0.0%	1,040
liabilities	0.8%	0.8%	0.0%	1,138
pension_policies	0.8%	0.8%	0.0%	538
assets	1.0%	1.0%	0.0%	8,435
households	0.3%	0.3%	0.0%	316
income_items	0.9%	0.9%	0.0%	1,040

Holdout test per-entity metrics

Entity	Missed	Errors	Invented	GT
people	0.3%	0.3%	0.0%	355
liabilities	0.8%	0.8%	0.0%	355
pension_policies	0.8%	0.8%	0.0%	170
assets	1.0%	1.0%	0.0%	2,723
households	0.3%	0.3%	0.0%	1,184
income_items	0.9%	0.9%	0.0%	355

Train F1 by extracted field, per household, for each dialog.

Holdout test F1 by extracted field, per household, for each dialog.

Quality controls

Each stage has its own checks before the next stage consumes its outputs

Quality is considered at a pipeline prospect, not only as a final accuracy number. We validate the generated data, the dialogs, the extraction outputs, the scoring outputs, the dialog-grounded GT, the extraction behavior on Train, and the full pipeline on the holdout test.

1. Synthetic data quality

The household generator is checked with business rules that enforce relationships between fields (married implies two adults, parents/child age gaps, mortgages/pension caps, aligned marital status compatibility). Distributions are reported alongside the generated data.

Statistical tests comparing the generated data against public-data anchors (Census/ACS, Fed SCF, and Jensen Huston divergence, Kennedy and Ormerod 2021) are planned but not implemented. Geocoding and external validation are out of scope for this synthetic-household artifact.

Rule violations	Distribution by persona	Anomaly scores
Automatic hard-constraint checks	Distribution of key variables by persona	Isolation Forest flags unusual rows

2. Dialog generation quality

Dialogs are generated in stages rather than as a single monolithic response: persona, outline, plan by phase/transcript, and then rendered. Validators and retries exist where the model has a measurable failure mode, such as missing required fields or invented facts.

Evidence anchors keep details in the transcript grounded in structured facts. These evidence fields later become dialog-grounded GT for extraction scoring.

Dialog-level checks	Validator-aware retry	Deep/Rock judge
Length, coverage, and required-field checks	Regenerate with validator feedback when a transcript fails	Classifies each transcript as realistic but simple or challenging

3. Ground truth quality

Ground truth is not treated as perfect. Dialogs can introduce ambiguity or omissions. This is expected. We separately track strict schema GT and dialog-grounded GT so that extraction errors can be interpreted as model errors rather than automatic proof that the transcript was wrong.

The GT-aspect reports how many dialogs were usable, how many fields were dropped, and how many records remain after filtering. Dialogs were scored using strict schema GT, dialog-grounded GT, and holdout evaluation, plus qualitative review of dialogs, validators, generated profiles, and, and an annotation guideline for ambiguous cases.

Strict schema GT	Dialog-grounded GT	Holdout evaluation
Strict schema target	Dialog-grounded target	Held-out quality check

4. Extraction quality on Train

The dev slicing, Train set, is used to develop the extractor, prompts, normalizations, record pairing, record matching, and error reporting. This is where we inspect failures, tune validators, and decide which changes are worth keeping.

Train quality is monitored through structured error reports, current-versus-previous comparisons, and per-scenario breakdowns. Some remaining errors are expected and are treated as the benchmark's difficult tail, not as evidence that the pipeline is broken.

Train extraction report	Current-run summary	Field-level errors
Main Train extraction output	Compares Train runs	Shows which fields still fail most often

5. Holdout quality

Holdout is used only after the Train pipeline is stable. It is the held-out evaluation for the full pipeline, including dialog generation, extraction, and scoring. Holdout results are reported as the benchmark's current held-out evaluation.

Holdout evaluation	Holdout scoring report
Main held-out evaluation	Scores the held-out extraction output

6. Pipeline quality on the holdout test

Holdout quality is reported at several levels: record-level F1, exact record rate, invented-record rate, missing-record rate, dialog-grounded GT coverage, extraction, overall matching, scoring, disagreement, and per-scenario breakdowns. Holdout results are treated as the current held-out benchmark for this synthetic-household dataset.

Holdout extraction report	Holdout scoring report	Holdout disagreement report	Holdout disagreement summary	Holdout scores table
Main held-out extraction output	Main held-out scoring output	Compares Train and holdout disagreement patterns	Holdout disagreement summary	Holdout scores table

Next steps

Research and product directions

The current benchmark gives us a controlled, end-to-end synthetic-data pipeline with explicit quality checks and a held-out evaluation. The next work is about harder conditions, and customer-extraction artifacts and product-facing outputs.

1. Improve dialog realism with multi-model critique

Dialog generation is sensitive to several model failures such as Jerkiness, Genericness, Inconsistency, Over-explanation, and Over-politeness. Each of these can degrade dialog quality, and some of them may also harm fields, extraction, issues, latency, and cost. We plan to use a multi-model critique and revision loop to improve dialog quality.

2. More noise-based stress tests

Inject controlled noise into dialogs and measure how extraction metrics degrade. Noise types include Transcription errors, Missing or ambiguous details, Speaker confusion, Inconsistent fact statements, Irrelevant side conversations, and Formatting inconsistencies.

3. Study dialog length versus extraction quality

Quantify how transcript length affects extraction quality, latency, and cost. Longer dialogs are a realistic stress test, but they may also introduce more distractors and more opportunities for hallucination.

4. Add a pre-enrichment extractor

Compare direct extraction against a two-stage variant that first extracts a compact fact sheet and then runs the main extractor. This may improve robustness on noisy, ambiguous, or rock-heavy transcripts, at the cost of an extra inference step.

5. Add adversarial dialog attacks

Test robustness against adversarial inputs, such as prompt-injection attempts that try to override the extractor, or transcripts that try to force invented fields or hallucinations. This is a stress-test extension, not a core benchmark objective.

6. Add a pre-enrichment extractor

Compare direct extraction against a two-stage variant that first extracts a compact fact sheet and then runs the main extractor. This may improve robustness on noisy, ambiguous, or rock-heavy transcripts, at the cost of an extra inference step.

7. Make infrastructure failure-tolerant

Add checkpoint/resume for API outages and rate limits, partial-run recovery, cleaner retryable vs. non-retryable errors, and a resumable, restart-safe pipeline. This is an infrastructure reliability and cost-control item, not a core extraction or dialog quality change.

Artifacts & reproduction

Useful local references

These links point to files in this repository and are useful for reviewing the pipeline, reports, and scripts.

Data generation

Households	Scenarios	Generation summary
`data/synthetic_households.jsonl`	`data/scenarios.jsonl`	`data/generation_summary.txt`

Dialogs

Train dialogs	Holdout dialogs	Dialog-grounded GT
`data/dialogs_train.jsonl`	`data/dialogs_holdout.jsonl`	`data/ground_truth_dialog.jsonl`

Extraction and results

Train extraction	Holdout extraction	Train scoring report
`extraction/runs/train/extracted.jsonl`	`extraction/runs/holdout/extracted.jsonl`	`scoring/reports/train/report.txt`

Holdout scoring report	Train disagreement report	Holdout disagreement report
`scoring/reports/holdout/report.txt`	`scoring/reports/train/disagreement.txt`	`scoring/reports/holdout/disagreement.txt`